

Single-Cyclist Crashes: A review of the current classifications and analyses

Lorem ipsum



Figure 1: Yes, that's me in a ditch with the sidecar bicycle

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1 Introduction

Cycling population is on arise due to the advantages of the bicycle as a commuter vehicle. This refers from almost free transportation and less pollution to healthier lifestyles and less traffic congestion. However, as the cycling population increases, their exposure to crashes also increase.

Regarding bicycle crashes, one configuration that stands out due to their major share on statistics are the Single-Cyclist Crashes (SCC) (Wegman and Schepers, 2024). These are events where no other road user is involved as an active perturbation, i.e. its effect does not depend on the source. Its share ranges from 50% to 85% of cyclist-related hospital admissions. Moreover, it is known that these events are commonly underreported Boele-Vos et al. (2017); Beck et al. (2019); Schnee et al. (2021); Utriainen (2020); Utriainen et al. (2023a).

In the last decade, Europe has seen a reduction in the fatality rate of total road accidents, except for bicycle crashes (CBS, 2024; ?; Gildea et al., 2024; Salmon et al., 2022; Utriainen et al., 2023a). Indeed, in the Netherlands the risk of death in SCCs has not only remained constant, but increased up to 7% (Schepers et al., 2017). Therefore, it is a matter of interest when bicycle usage is intended to be promoted and risks need to be minimised. Furthermore, if these issues are not attended, bicycle usage can be expected to decrease (Gu et al., 2019). However, the dynamics of these crashes are not well understood Haworth and Debnath (2013). Therefore, research on the dynamics of SCCs is needed to continue developing cyclists road safety.

Up to this point, most work has been done in descriptive analysis of common factors that lead to SCCs (Beck et al., 2019; Gildea et al., 2024; Hertach et al., 2018; Salmon et al., 2022; ?; Schepers and Wolt, 2012; Schepers et al., 2015; Utriainen et al., 2023a). Within these descriptive analyses we find common factors and mechanism with a major role in the crash. However, none of these classifications take into account the particular dynamics of the bicycle. Therefore, the aim of this article is to review current classifications and develop a bicycle dynamics based classification of SCCs.

The importance of provide a classification based on bicycle dynamics is to highlight the challenges that are needed to overcome to continue developing the safety of the bicycles. Also to better understand what are the key factors that casuse bicycle crashes and propose more effective countermeasures.

1.1 Key concepts

Since some concepts in this matter are not standarised, we include important definitions used in this work.

First, the events of interest receive different names: single-bicycle crash, single-actor bicycle crash, single-cyclist crash, among others (see Table 3). On one hand, the expresion single-bicycle crash neglects the rider, making reference to a problem only oriented on the bicycle. On the other hand, single-actor bicycle crashes assumes that there are no other actors involved in the event, which neglects indirect effects of other road users, e.g. pedestrian entering into the bike lane and triggering an avoidance manoeuvre that ends in a crash. In addition, any other denomination that includes the word ‘accident’ is not appropriate due to the preventable nature of these events (Davis and Pless, 2001). Therefore, for this work we adopt single-cyclist crash, due to its reference to the rider and the bicycle as the main actors in the event, allowing other indirect actors.

Second, during this article, we will refer to the bicycle and the human coupled together in motion as a dynamical system, which is modelled by non-linear differential equations. Assuming the rider as a controller of the system, we identify the existence of *controllability*, which means that all system variables can be influenced by the input (Seffen et al., 2001; Verlinden and Kabeya, 2012).

Third, a perturbation will be understood by its dictionary definition as any variation in the system that causes it to change its regular movement (University, 2025). An active perturbation is a controlled force from an external source, such as other vehicle colliding with the system. A passive perturbation is a non-controlled force, e.g. the force exerted by road irregularities on bicycle tyres.

Fourth, we will call a crash to any sudden and unintended event that results in a disruption of the normal and controlled motion of the human-bicycle system. In the analysis of vehicular collisions, there are three phases: pre-impact motion, impact motion and post-impact motion (Park and Han, 2001). Translating this approach to bicycle crashes, pre-crash motion encompasses the time window from the perturbation to the moment where the bicycle is no longer controllable. Then, the crash stage begins, until the moment when the system stops its motion. Post-crash stage is limited in SCCs and can be assessed as the moment when the vehicle and the rider are disconnected and presenting independent motions (Lloyd, 2016).

Fifth, as we are dividing the events based on objective and defined criteria, the result is a classification. Although both concepts are used as synonyms, categorization is used for more creative and flexible divisions ?.

At last, bicycles and motorcycles share characteristics as a dynamical systems, which makes them similar for studies. In its analytical study, both are usually assumed as a four-bodies single-track vehicle Sharp (1971); Meijaard et al. (2007); Lot and Sadauckas (2021), with two basic degrees of freedom: roll φ and steering δ . Both vehicles experience similar crash configurations (Haworth and Debnath, 2013; ?; ?), including tyre sliding, inability of control the system through the steering, excessive pitch rotation, among others later discussed in this paper. The main differences are the mass, the working speed range, and power input from the user, being

this last what defines if the vehicle is a bicycle or a motorcycle. Despite some modern electric bicycles include advanced suspension systems, heavy batteries and electric motors that increase overall mass, and are capable of reaching speeds up to 45 [km/h], they are still dependant on rider force for longitudinal motion.

1.2 Research target

The questions that shape the present article are:

1. What are the common approaches to study and analyse human-bicycle dynamics and control in crash scenarios?
2. Is there a classification of crashes that captures the particular behaviour of the human-bicycle system?
3. Is there single-cyclist crash data that is useful to study the dynamics of these events?

The article is structured as follows, Section ?? contains the basic anatomy of a crash, explaining the required factors to occur. Additionally, we review current work on classifications and SCC characteristics, along with the main concepts on the field of bicycle dynamics. The proposed classification is developed on Section ?. Finally, Sections 5 and 6 present the discussion and conclusion, respectively.

2 Methods

The literature search was made in Google Scholar and Dimensions database in October 2024. * The interest relied on studies that analysed the dynamics of single-track vehicle crashes, which include bicycles, motorcycles and scooters. Therefore, the initial query for the broad range search on October 2024 was:

(bicycle OR bike OR two-wheel* OR single-track OR “two wheel*” OR “single track”) AND (crash OR collision OR accident OR fall).

The results of the search provided the main articles in the field, however, we used ‘snowballing’ methodology to complement the search. This approach was particularly useful to cover base concepts that are not always clarified.

2.1 Inclusion and exclusion criteria

The inclusion criteria for the snowballing technique was:

- The study presented statistics related to single-cyclist crashes.
- The study takes into account the dynamics of the bicycle for the crash analysis.
- Studies concerning human control of vehicles.
- Studies that explain the techniques to control a bicycle.
- Analyses on bicycle dynamics that explain its rideability and controllability.

On the other hand, the exclusion of articles was based on the following criteria:

- Studies focused only on injury outcomes.
- Analyses of crashes of bicycles with other road users.
- Articles in a language different than English.

2.1.1 Some remarks on crashes with other road users

Bicycle crashes can be clustered in four categories like Gildea et al. (2024), classifying them according to the nature of the other road user. In these categories, we find bicycle-bicycle, bicycle-motorised vehicle, bicycle-pedestrian and single-cyclist crashes. Categories including other road users are out of the scope of this review since those perturbations act as an active, uncontrollable and unpredictable perturbation on the human-bicycle system.

3 Results

- Schepers classification -¿ what does it miss? - Crash motions -¿ Gildea, Rose and Petit. - Mechanisms?

In the most basic understanding of a single-cyclist crash, this event is the result of the rider losing control of the vehicle. Fuller (2005) describes the loss of control as a relation between task demand and rider capability, when the task demand surpasses the rider capability, the rider loses control and the crash occurs.

Fuller (2005) presents the task-capability model for car drivers, which can be extrapolated to bicycle riders. This model considers physical characteristics and knowledge of the driver on what to do under determined circumstances to define the boundaries its capability. However, this capacity is also affected by other factors related to the circumstance, such as awareness, emotion, fatigue, among others. On the other hand, task demanding elements encompass travelling speed, environmental factors, such as road characteristics and other road users. Of these factors, the driver has direct control of the travelling speed, achieving balance between driver capacity and its awareness to the task. While car drivers can reduce task demand by decreasing speed, this is not always the case for bicycle riders, due to the increase of user stabilisation requirements at low speeds Kooijman et al. (2011); Schwab and Meijaard (2013); Vlakveld et al. (2015). In general, the theory shows an almost constant driver capability balancing with changing task demands, which should not surpass the capability threshold.

When the task demand increases exponentially, possibly leading to a loss of control, it is understood as a critical scenario . Critical scenarios and factors that contribute to them are known in the literature. As seen in Table 3, the most repeated events are sliding, low speed riding, braking, loss of control, collision with an object and interactions with tram tracks.

Author	Country	Data origin	Share of SCC ¹	Classification	Main type (%)	Name
Algurén and Rizzi (2022)	Sweden	-	48 - 95% ²	Crash characteristics	Skidding (26%)	SBC ⁷
Beck et al. (2019)	Australia	Hospital data	48%	Mechanisms	Avoidance braking	SCO ⁸
Boele-Vos et al. (2017)	Netherlands	Ambulance and police reports	80%	Crash type	Object collision	SBC
Boufous et al. (2013)	-	Hospital and police data	>50% ²	Collision/No-collision	Loss of control (80%)	SVBC ⁹
Gildea and Simms (2021)	Ireland	Self-made survey	29%	Causes and factors	Slippery roads	SCC ¹⁰
Hertach et al. (2018)	Switzerland	Survey	-	Mechanisms and factors	Skidding (31%)	SVC ¹¹
Meulenens et al. (2020)	-	Hospital data	39%	-	-	SCC
Ohlin et al. (2019)	Sweden	Traffic database	68% ³	Characteristics	Skidding (14%)	SBC
Schepers and Wolt (2012)	Netherlands	-	-	Mechanisms	Riding off the road (21%)	SBC
Schepers et al. (2015)	International	-	60 - 95% ²	-	-	SBC
Schepers et al. (2017)	Netherlands	Police report	-	-	-	SBC
Schwab et al. (2018)	Netherlands	-	68% ⁴ / 77% ⁵	-	-	SBC
Utriainen (2020)	Finland	Insurance data	-	Mechanisms	Skidding (47%)	SBC
Utriainen et al. (2023b)	-	-	52 - 85%	-	Loss of control or skidding	SBC
?	Finland	Crash institute data	35.2% ⁶	Factors and mechanisms	Riding off the road (71.9%)	SBC
Vanparijs et al. (2016)	-	Insurance data	31%	Human factors	Riding off the road (21%)	SBC

¹ Of total bicycle crashes.

² Of total cyclist hospital admission.

³ Of traffic crash database.

⁴ Normal bicycles.

⁵ Electric bicycles.

⁶ Of fatal bicycle crashes.

⁷ Single-Bicycle Crash.

⁸ Single-Cyclist Only.

⁹ Single-Vehicle Bicycle Crash.

¹⁰ Single-Cyclist Crash.

¹¹ Single-Vehicle Crash.

Literature with statistics about single-cyclist crashes.

3.1 Single-cyclist crash classifications

- On the literature we find 3 main classifications: Schepers, Gildea and Schnee.

- Schepers considers infrastructure, rider, bicycle and other factors with their respective subclassifications. - This approach is good but lack of information required for reconstruction and analysis. - Seems to be centered in the factors more than in the real mechanisms. - Puts bag on the handlebar as rider fault, when it should be assumed as an external perturbation. - In accordance between both researchers that did the publication.

- Gildea uses the multilayer approach. - Based on factors that cause the crash: slippery roads, tram tracks, kerb collision, avoidance manoeuvres and collision with potholes.

- Schnee focuses it on detecting crashes through IMUs - Parameters: State of riding, collision, ground hit and tipping over. - This needs definitions for the sub classes.

- Gildea 2023 Computer talks about sub-division of Loss of control: minor, substantial or complete.

Read comments on Daily/20w25 for more criteria here.

In this point, we encounter the lack of information about the dynamics of the accidents. Across the studies, the conclusion is that the most frequent causes of SBCs are skidding, loss of control by braking or steering manoeuvres, and external perturbations such as hitting a kerb (Schepers and Wolt, 2012; Hertach et al., 2018; Utriainen, 2020; Algurén and Rizzi, 2022). Moreover, the way of interaction with the environment can increase the risk, e.g. the angle of crossing a tram line (Gildea et al., 2023). However, there is no information about which one of the wheels slips, longitudinal or lateral velocity, steering angle, braking force, etc. Therefore, it is necessary to gather data to assess bicycle crash dynamics and propose countermeasures from the human-bicycle system itself.

From the factors mentioned, it is possible to identify the main mechanisms that trigger the critical situation. When the front wheel slips, the main control input from the rider disappears, reducing the system to an inverted pendulum without user input. Events such as getting into tram tracks, lock the steering, first eliminating one of the intrinsic characteristics of bicycle's self-stability (Kooijman et al., 2011) and then, the rider is unable to input any control action. Collisions in the front assembly (front wheel, fork and handlebar) generate a sudden steering input. Bicycle malfunction such as chain slip can lead to rider disengagement. At last, avoidance manoeuvres initially occurs in a open-loop control action (Nugent et al., 2019), which leads to excessive roll and steering rates, or in a few cases, it leads to pitch over.

3.2 Bicycle dynamics

Bicycles fit into the category of single-track vehicles due to their

A bicycle is a **human-powered single-track vehicle**, which behaviour has been a matter of scientific interest for more than 100 years. The now classical approach for its study considers the bicycle as a dynamical system composed of four rigid bodies: rear frame, front frame, and two wheels.

Assuming a rider rigidly attached to the bicycle, Meijaard et al. (2007) presents the linearised equations of motion for the three degree of freedom model. From these linear equations, the stability of the system is studied, showing two main modes of vibration: Weave and Capsize, which are the particular interest due to its danger to the rider when unstable. Where weave is an oscillatory motion, from roll and steering, while capsizes is a non-oscillatory inverted pendulum-like motion. Thus, the linearised bicycle model was shown to be stable in the range of 4.3 to 6.0 $[\text{m s}^{-1}]$. Furthermore, if tyre parameters are included, wobble mode arises (Sharp, 1971; Schwab and Meijaard, 2013), which is an oscillatory motion of the steering, also danger when unstable.

(Kooijman et al., 2011) demonstrated that the basis of bicycles' self-stability relies on mass distribution and steering coupled to lean angle. These results showed that self-stability is more than angular momentum of the wheels and geometric trail distance as was commonly assumed.

Once understood the basic behaviour of the bicycle itself as a dynamical system, the intuitive next step is the control of the bicycle: How do we ride a bicycle? The task of riding a bicycle can be simplified to maintaining system stability and following a desired trajectory. On the other hand, maintaining system stability has been assessed to understand how bicycles are controlled. In this field, it is known that most control actions come from the steering, and at low speeds knees start to play a role in the control task (Kooijman et al., 2009).

Then, research has focused in developing a human controller to model rider behaviour (Schwab and Meijaard, 2013; Kooijman and Schwab, 2013). An aircraft pilot control simplified model was presented by Hess et al. (2012). Additionally, Wang et al. (2020) develops a controller which takes into account time delays of human body and compared the results with experimental data. Equally, Dialynas et al. (2023) takes into account steering torque feedback and sensory delays to develop his rider model.

The next step, after understanding the basics of human control, is to assess the handling of the bicycle. In particular, Ronné et al. (2024) defines it as '*...the ease and precision with which a human can perform a given control task*'. With this approach, the analysis enters into the concept of Task Demand (TD) and Rider Capability (RC) (Fuller, 2005). As a result, when TD increases over RC, the rider loses control of the system and enters in the crash scenario.

3.3 Active Safety Systems

The interest of including current active safety systems in the present review is to analyse the way they detect a critical scenario and what is the response provided to minimise risks.

3.3.1 Antilock Braking System (ABS)

This system consists in a constant reading of vehicle's wheels speed and it is capable of detecting when a wheel locks, decreasing the braking pressure and maintain the wheel spinning while diminishing the velocity. The main result in four-wheeled vehicles have been the possibility of maintain the control of the vehicle during the whole braking manoeuvre. Concerning motorcycles, it prevents low-side and pitch-over crashes by avoiding front wheel lock and high-side crashes by avoiding rear wheel lock. It was discovered that motorcycles with ABS system reduce in more than 20% on average the involvement in crashes (Rizzi et al., 2013). Similarly, (Teoh, 2022) found a 22% reduction of fatal crashes per year with ABS motorcycles.

4 Bicycle dynamics-based crash classification

Linear crash causality models are widely available and already used, however, towards the aim of Zero fatalities, new models are required that consider different combination of factors to provide a more integral analysis Zhang et al. (2022).

The proposed classification is a multi-layer classification, which allows to combine the influence of different factors on the outcome of the event. The classification consists of four main layers and three sub-layers, as seen in Figure 2. The main layers are made of already known concepts in the field, such as manoeuvres, excitation of the system, mechanisms of crash, and motion of the system during the crash. On the other hand, sub-layers are composed of found gaps in the literature, such as human control in crash or pre-crash scenarios, characteristics of the excitation of the system and the location of the mechanism.

The multi-layer approach allows to classify the crash event with different depth levels according to the available information. Thus, this gives a flexible base that can be applied in different fields, depending on the objective sought.

The way of understanding the diagram on Figure 2 goes: The rider is executing a manoeuvre, the human-bicycle system experiences a excitation that leads to a critical situation. Depending on the human control and the characteristics of the signal, the task demand increases above rider capability limit and a crash mechanism acts. From the crash mechanisms, we can identify its main location, which leads to one of the characteristic motions, to finally end in a crash.

4.1 Manoeuvre

The first layer consists on the manoeuvre that the rider is executing at the moment of entering to the critical scenario. Here we understand straight-running as the action of riding a straight path, maintaining balance through minor steering adjustments and no major variations on the states of the system. Longitudinal acceleration manoeuvres refer to braking and acceleration, both performed by the user. Steady and planned cornering refers to the situation where the rider is performing a controlled turning. Low speed makes reference mainly to start and stop riding, where users require major balance actions to maintain the bicycle in upright position. Unplanned turning or avoidance manoeuvre is when the rider executes a sudden evasive turn to avoid an obstacle.

Closing paragraph

4.2 Excitation of the system

Due to the complexity of the human-bicycle system, it is subjected to several excitations:

- **Intrinsic (Int):** Makes reference to any excitation that occurs due to the behaviour of the bicycle itself. Mainly composed of the modes of vibration.
- **Rider (Rid):** Rider excitations are linked to rider control inputs, if they are incorrect, it leads to a failure mechanism.
- **External (Ext):** This refers to any perturbation, force or change of state exerted by the environment to the system.

Closing paragraph

4.3 Mechanisms of crash

Currently on the literature, we find several known mechanisms of crashes, normally clustered as skidding, collision with obstacle, falls, braking, drifting off the road, among others. However, these ‘boxes’ do not fit the broad spectrum of mechanisms, nor capture the dynamics involved in the events. For this reason, we propose the following mechanisms based on the literature and observation from a dataset of publicly available videos .

Since the aim of this classification is to capture the behaviour of the bicycle, there is no difference between the reason why tyre friction reaches its saturation limit. Thus, sliding on wet road, wet leaves, ice or gravel can be simply clustered into tyre friction failure.

- **Modes of vibration:** Oscillatory modes of vibration of the bicycle: weave and wobble.
- **Rider-bicycle joint:** The rider is ‘attached’ to the bicycle in 5 points: handlebar (both hands), pedals and seat. When one or more of these deataches, the rider loses a control input to the system.
- **Balance control:** This makes reference to the ability of the human of balancing the bicycle as an inverted pendulum, moving the steering to control its equilibrium. When the control action is not correct, the rider reaches a point where is not possible to maintain the balance. An edge-case example are bags hanging on the handlebar, these affect the steering leading to a failure on the balance control.
- **Load transfer:** Longitudinal accelerations generate longitudinal load transfer, which remains unnoticed for most casual riders. However, under large accelerations, this load transfer exceeds the maximum tolerable values for maintaining the bicycle in a stable configuratiopn.
- **Tyre friction:** The only connection between the bicycle and the environment is on the tyre-ground contact interface. Tyres have a saturation limit for the grip offered, when this limit is reached, they start to slide and the rider loses control of the vehicle.
- **Contact forces:** This makes reference to any collision or force exerted on the bicycle, such as collision with an object, or steering perturbations due to uneven road surfaces.
- **Aerodynamic forces:** Aerodynamics can play an imporatnt role when large wind gusts strike the rider by the side, generating a major lateral force, which can destabilise the system.

On the literature, these mechanisms and their approaches are treated as absolutes, with no distinction on the application point. For this reason, we include front or rear, along with longitudinal or lateral features to complement the mechanism.

Closing paragraph.

4.4 Characteristic motion

Currently on the literature we find three characteristic motions for two-wheeled single-track vehicle crashes ?, see Figure ?? for schematic representation of the motions.

- **Pitch-over:** The main characteristic of this motion is one of the wheels lifting from the ground, following a trajectory that finishes with the front wheel behind the rear wheel.
- **Low-side:** The human-bicycle system follows an excessive roll rate in the same direction as at the beginning.
- **High-side:** Characterised for a sudden deceleration of the wheel while in lateral motion, which leads to a violent negative roll rate with respect to the direction of the initial motion.

However, it is possible that the crash scenario does not provide a characteristic motion of the bicycle. In those cases, it is possible to assess it from the characteristic motion of the rider on the event.

5 Discussion

6 Conclusion

Acknowledgement

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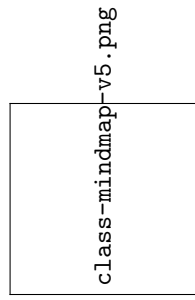


Figure 2: Multi-layer single-cyclist crash classification. Solid lines represent already identified and known concepts in the field. Dashed lines represent gaps that are currently not discussed in the literature.